DAY 39 · THEORY HANDOUT

# Let an Agent Search Documents

Wrap retrieval as a tool and check when the agent uses it.

KGP Talkie

Scan to subscribe



Laxmi Kant Tiwari · kgptalkie.com

# The idea and its useful terms



Wrap retrieval as a tool and check when the agent uses it.

### Retrieval tool

A tool that returns matching document text.

### Agentic RAG

An agent decides when to retrieve evidence.

### Tool trace

The recorded tool calls and returned results.

Standard RAG retrieves on every question. Agentic RAG can choose whether a search is needed and can ask a revised search question.

That choice adds flexibility and another place for mistakes. Check that document questions actually cause a retrieval call.

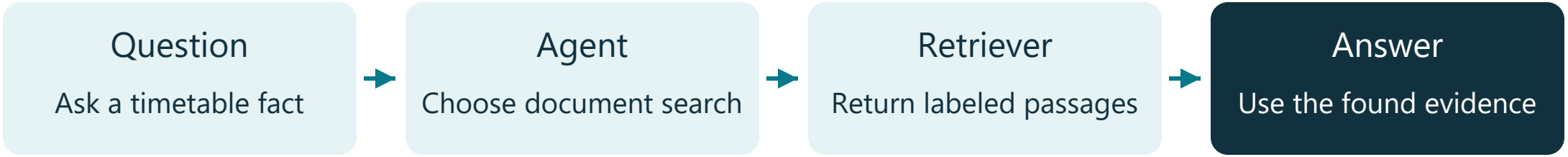
> Agentic RAG can skip a needed search. The system instruction helps, but the actual trace tells us what happened.

# Follow the work, one step at a time



Each step has a job. We can inspect its output before moving on.

Read the path from left to right.




Each arrow means: pass the result to the next step.

Wrap the retriever as a tool with a clear description. The agent can request it using a search question. The tool returns passages with source labels, and the agent writes a reply from those results.

A revised search can help after a poor first match, but it costs another step. A greeting needs no timetable fact. A document question does need evidence, so check the trace instead of assuming retrieval happened.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

# A small example we can check



The values here are classroom examples. The notebook shows actual results.

### Greeting

Hello.

No document fact needed.

### Course question

When does Pandas start?

Search the timetable.

### Missing fact

How much is the fee?

Search, then say unknown.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> Agentic RAG can skip a needed search. The system instruction helps, but the actual trace tells us what happened.

05 KGP Talkie

# Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

## Fixed RAG

Fewer moving steps.

Good for narrow Q&A.

## Agentic RAG

Tool choice can vary.

Inspect each search.

## Evidence rule

Use retrieved facts only.

Keep their labels.

Build the index inside this notebook so it does not depend on another day's files.

Turn the retriever into one typed tool and create an agent.

> Test a greeting, a supported question, and a missing fact. Check the tool trace for each.

# Practice, check, and read the full lesson



Use the notebook to try the steps. Use the article when you need the longer explanation.

### 1. Build

Run the small example.

Inspect each step.

### 2. Change

Try the fresh input.

Keep the goal clear.

### 3. Check

Compare with the source.

Keep mistakes visible.

Test a greeting, a supported question, and a missing fact. Check the tool trace for each.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> Agentic RAG can skip a needed search. The system instruction helps, but the actual trace tells us what happened.

DAY 39 · COMPLETE

# Thank You

Follow KGP Talkie for new lessons, course updates and live sessions.

Scan to follow

**Instagram**
@kgptalkie

**WhatsApp Channel**
KGP Talkie